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Robotic-supported Data Loss Detection in Android Applications

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Smartphones have become integral to modern life due to their diversity. However, the development of applications for smartphones presents significant challenges, primarily due to two factors: (i) the diversity of hardware and operating system versions across multiple configurations; and (ii) the impossibility of testing the smartphone in a non-invasive way (without user interaction), which is expensive, tedious, and time-consuming as it is usually carried out manually. In this context, automation has emerged as an effective solution for the testing of diversity of smartphone systems and versions, and for performing testing tasks non-invasively. In addition, automation facilitates quick, accurate, and repetitive testing, thus enabling developers to validate their applications across various configurations effectively. This results in a less invasive verification process, a significant reduction in time spent on manual testing, thus accelerating the development cycle. Our work proposes R-DLD (Robotic-supported Data Loss Detection), a robot-assisted infrastructure for data loss detection in Android applications, offering more realistic tests by interacting directly with the devices. The robot is constructed using cost-effective components, facilitating its adoption in testing environments. In our evaluation, R-DLD successfully identified 341 data loss instances across 10 randomly selected apps from an Android store. All identified issues were received responses from the developer, with 89.55% of the issues being resolved, while 35.82% have been subsequently resolved.

Keywords: automated robots, testing, android, non-invasive testing

1 INTRODUCTION

Smartphones have become an essential tool in people's lives due to their ease of use, and the availability of applications that enhance daily life in various dimensions (work, shopping, education and communication). However, the development of quality applications for smartphones presents significant challenges, primarily due to two factors: (i) the diversity of hardware and operating system versions across multiple configurations; and (ii) the impossibility of testing the smartphone in a non-invasive way (without user interaction), which is expensive, tedious, and time-consuming as it is usually carried out manually. In this context, automation has emerged as an effective solution for the testing of diversity of smartphone systems and versions, and for performing testing tasks non-invasively. In addition, automation facilitates quick, accurate, and repetitive testing, thus enabling developers to validate their applications across various configurations effectively. This results in a less invasive verification process, a significant reduction in time spent on manual testing, thus accelerating the development cycle. Our work proposes R-DLD (Robotic-supported Data Loss Detection), a robot-assisted infrastructure for data loss detection in Android applications, offering more realistic tests by interacting directly with the devices. The robot is constructed using cost-effective components, facilitating its adoption in testing environments. In our evaluation, R-DLD successfully identified 341 data loss instances across 10 randomly selected apps from an Android store. All identified issues were received responses from the developer, with 89.55% of the issues being resolved, while 35.82% have been subsequently resolved.

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In this context, one promising solution is the use of robotics in mobile device testing. Robotics offers a form of black-box testing that has not been fully "witnessed" thus far. Using a robotic arm to interact with and mobile device interaction, the source code of the application is not required.

Recently, Riganelli and others [1] proposed a Data Loss Detector (DLD), a technique for detecting data loss in Android applications, data that is accidentally deleted or when the application returns to default or initial values. This is a critical issue in the life cycle of the Application, as it can lead to the loss of data for implementing the application. Depending on the state received, the Activity can be restarted, and resources can be later released. Events, which occur frequently in the application, for example, a rotation of the screen from portrait to landscape [2].

Inspired by DLD, and motivated by the need for testing devices in an environment that simulates real-world conditions, this paper introduces a data loss detection in Android applications supported by a Robotic-supported Data Loss Detector. The detector uses sensor inputs to induce device states, thus enabling the detection of data loss in a less invasive and more efficient manner.